

Task knowledge overlap and knowledge variety: the role of advice network structures and impact on group effectiveness

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Summary

Drawing from the structural perspective of social capital theory, this research investigates how internal and external advice network structures influence knowledge overlap and variety and, how these knowledge dimensions in turn influence group effectiveness. Findings from two studies on knowledge-intensive groups indicate that different advice network structures are associated with knowledge overlap and knowledge variety, and only knowledge variety was significantly associated with group effectiveness. In addition, despite implicit understanding that advice networks aid performance through enhancing knowledge outcomes, only knowledge variety was found to mediate the relationship between external network and group effectiveness. Implications for theory and practice are discussed. Copyright © 2007 John Wiley & Sons, Ltd.

Introduction

Groups involved in knowledge-intensive work typically have to acquire and leverage a wide range of task knowledge such as marketing, technological, or financial knowledge so that they can understand the complexity inherent in their tasks. Yet, group members are also likely to need shared task knowledge in order to communicate and coordinate effectively to execute the group task. Not surprisingly, studies in new product development teams and experimental groups involved in complex cognitive tasks have shown that *knowledge variety*—defined as coverage or range of different types of task knowledge in a group—and *knowledge overlap*—defined as the extent to which group members have common task knowledge—are significant for group effectiveness (Hoopes & Postrel, 1999; Rulke & Galaskiewicz, 2000; Walsh, Henderson, & Deighton, 1988).

Traditionally, as research in the group heterogeneity tradition has tended to collapse knowledge variety and overlap into a ‘cognitive diversity’ construct (e.g., high cognitive diversity means high variety and low overlap), the implication is that high variety comes at the expense of overlap and vice-versa. However, knowledge variety captures the overall breadth of knowledge domains

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Received 17 October 2006

Revised 27 June 2007

Accepted 3 July 2007

represented in a group (e.g., Lyles & Schwenk, 1992; Walsh et al., 1988), and knowledge overlap indicates the extent to which, out of this range, members share common knowledge (e.g., Hoopes & Postrel, 1999; Mathieu, Heffner, Goodwin, Salas, & Cannon-Bowers, 2000). Hence, these two dimensions are conceptually distinct and can simultaneously co-occur. For instance, a group of specialists in diverse knowledge domains will have high variety but low overlap, but a group of generalists with the same coverage of knowledge domains will have both high variety and overlap.

Hence, a question of both theoretical and practical interest is how a group involved in knowledge-intensive work can optimize both knowledge overlap and variety. This paper offers a social capital perspective as one way of examining how these two knowledge dimensions can both be fostered in groups. The core premise of social capital theory is that social networks are conduits of resources (Burt, 2000; Granovetter, 1983). At the group level, social capital has been conceived as the configuration of group members' social relationships—both within and external to the group—through which necessary resources for the group can be accessed (Oh, Chung, & Labianca, 2004). Knowledge, in particular, has often been cited as a key resource that can be accessed through social ties and, then, used to improve group performance (e.g., Hansen, 1999; Reagans & McEvily, 2003). Focusing on the structural properties of social networks, studies on small groups have suggested that the internal and external social network structures of groups affect their performance through influencing their access to knowledge (e.g., Cummings, 2004; Reagans & Zuckerman, 2001).

Thus, drawing from the structural perspective of social capital, the main purpose of this paper is to propose and examine how internal and external advice network structures influence knowledge overlap and variety and, how these knowledge dimensions in turn influence group effectiveness of knowledge-intensive groups. The focus on task advice networks, as opposed to friendship networks, is grounded on the fact that advice networks are conduits for work-relevant information and knowledge and, thus, are likely to be the most relevant to predict group knowledge (e.g., Reagans & McEvily, 2003; Sparrowe, Liden, Wayne, & Kraimer, 2001).

This social structural network model of group knowledge is intended to advance both group knowledge research and group social capital research in several ways. For group knowledge research, the social network approach offers a theoretical perspective to understand how groups can leverage their internal and external advice ties to foster both knowledge variety and overlap at the same time. In addition, by disentangling the independent effects of knowledge variety and knowledge overlap on group effectiveness, this paper enables greater insights into which dimension is the driving force in affecting group effectiveness. This extends current group knowledge research that has, because of theoretical focus or experimental design limitations, addressed the effect of only one task knowledge dimension on group effectiveness (e.g., Hoopes & Postrel, 1999; McNamara, Luce, & Tompson, 2002; Rulke & Galaskiewicz, 2000; Stasser & Titus, 1985). From the group social capital perspective, while extant research has highlighted knowledge benefits as the key mechanism through which network structures aid group performance (e.g., Cummings, 2004; Hansen, 1999; Reagans & Zuckerman, 2001), the mediating effect of group knowledge on the relationship between network structure and group performance is inferred rather than examined directly. By tapping more directly into the implicit associations among network structures, group knowledge, and group effectiveness, the model in this paper also enables a more in-depth understanding of not only how different configurations of advice ties may lead to different group knowledge dimensions but also whether group knowledge is the causal mechanism through which advice structures influence group effectiveness.

The social structural network model of group knowledge, presented in Figure 1, is tested using two studies of groups involved in knowledge-intensive tasks. Study 1 examines whether advice networks and group knowledge at Time 1 affect subsequent group performance at Time 2 in a sample of 80 student groups involved in a complex business simulation task. Study 2 was conducted on 40 knowledge-intensive work groups from three organizations. For the purposes of this study, work groups

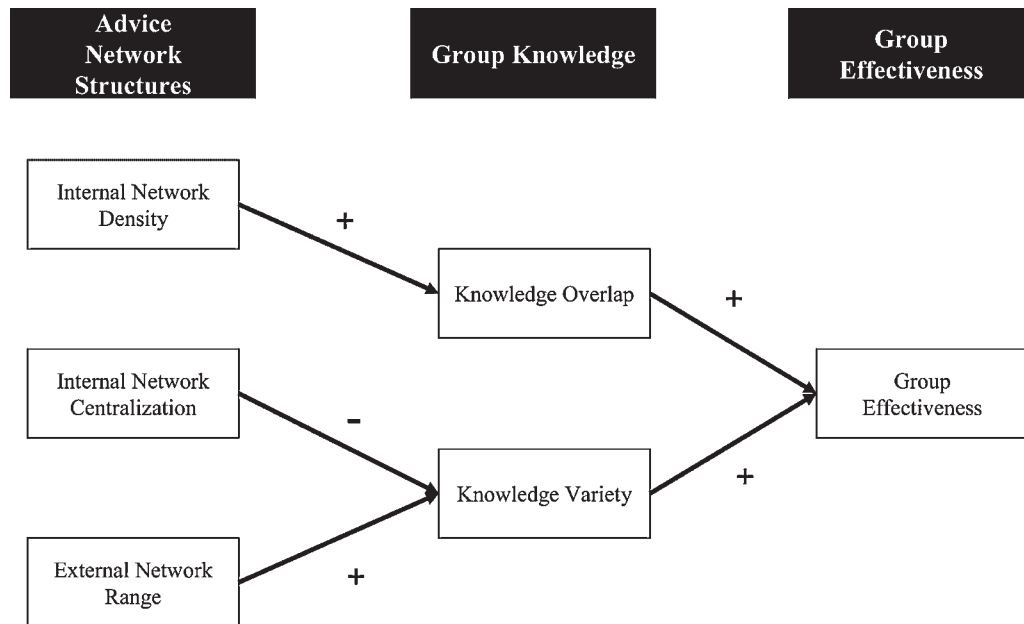


Figure 1. Hypothesized theoretical framework

are defined as groups of individuals in an organization with clearly defined membership, who are responsible for a shared product or service (Hackman, 1987).

Theoretical Background and Hypotheses

The structural perspective of social capital focuses on how the configuration of social ties facilitates the movement of needed resources, thereby enhancing performance (Burt, 2000). Because the overall pattern of social connections of a group can determine the extent of resource access, this approach has been valuable in predicting differences in group performance (Cummings, 2004; Oh et al., 2004; Reagans & Zuckerman, 2001; Sparrowe et al., 2001).

One class of explanations for how social network structures influence performance is grounded in how the overall configuration of ties facilitates knowledge creation and knowledge transfer. Nahapiet and Ghoshal (1998), for instance, posit that social network structures are likely to influence knowledge creation because they determine individuals' opportunity to access and combine knowledge. In a related vein, Reagans and McEvily (2003) found that network structure is associated with the ease of knowledge transfer between individuals because it is likely to affect not only the willingness of individuals to transfer their knowledge but also their ability to convey complex ideas more effectively to other people. In essence, these studies suggest that network structures are likely to influence knowledge transfer and creation processes in groups.

Drawing from this theoretical logic, this study proposes that both the internal and external advice network structures of groups affect the degree of task knowledge variety and overlap in a group. To the extent that advice network structures can influence the transfer and creation of knowledge (e.g.,

Nahapiet & Ghoshal, 1998; Reagans & McEvily, 2003), they are likely to affect the degree of knowledge overlap and variety in a group.

Advice network influences on knowledge overlap and knowledge variety

As knowledge overlap represents the degree of shared task knowledge among group members, it is likely to vary with the extent of internal knowledge transfers in a group. The more knowledge is transferred within a group, the higher the knowledge overlap. Internal advice network density in a group indicates the extent to which members seek task advice from one another and, thus, is measured by the average number of advice ties per group member. The higher the internal advice network density, the more each group member seeks advice from other members in the group. Because of the greater exchange of information and ideas among members, more individual knowledge is transferred within the group, thereby increasing the degree of shared knowledge. Direct interaction between individuals, for instance, is likely to facilitate the transfer of knowledge because there are greater opportunities to probe, clarify, and understand the information, thus enabling the transfer of knowledge from a source to a recipient (e.g., Hansen, 1999). Hence, when members are interconnected through advice ties, they not only have greater opportunity to access each member's knowledge but also the direct interaction should facilitate knowledge transfer among members, thus creating greater knowledge overlap.

Hypothesis 1. The higher the density of the internal advice network, the higher the degree of knowledge overlap in a group.

While internal transfers of knowledge increase knowledge overlap in a group, they are not likely to enhance knowledge variety because internal transfers merely move existing knowledge among group members but do not introduce new knowledge into the group. Rather, knowledge variety is dependent on the creation of new knowledge, which can be increased either through discovery of new understandings within the group or imported from outside of the group. Within the group, knowledge variety can increase when group members combine their knowledge to create new knowledge (e.g., Hargadon, 1999; Sutton & Hargadon, 1996), or decrease when group members exploit and converge on a limited set of knowledge domains (e.g., March, 1991). In this light, the ability of group members to mutually acknowledge and be receptive to one another's knowledge becomes critical because new understandings are likely to be created when members integrate different knowledge.

Increasing mutual respect and receptivity to different knowledge is likely to depend on the equality of member influence within a group. As studies of group member influence have shown, members can exert varying degrees of influence such that the opinions of the more influential members dominate the position of the group and those of the less influential are marginalized (e.g., Foushee, 1984; Kirchler & Davis, 1986). Internal advice network centralization, in particular, can foster inequality of member influence when there are increasingly a few group members who are the objects of advice-seeking from other members. Network centralization refers to the variance in network ties per group member, such that as network centralization increases, there are some members who enjoy proportionally more ties than other members, and subsequently become more central and influential in the network (Brass & Burkhardt, 1992; Sparrowe et al., 2001). Thus, in a highly centralized internal advice network, there are a few individuals who are most central in providing task knowledge. As such, we can expect their knowledge to become increasingly valued relative to others and they become increasingly influential in decision making (e.g., Bottger, 1984; Wittenbaum, 1998). This inequality in influence can lead to increasing deference to the knowledge of more central members (Kirchler & Davis, 1986). When this happens, the opportunity to create new understandings through integrating different viewpoints can be reduced as members become less likely to contradict the perspectives of more central members. In

addition, as the knowledge of more central members becomes more valued relative to others, there is the risk of convergence on these 'valued' knowledge domains and less emphasis on developing other knowledge domains, thus decreasing knowledge variety over time.

Hypothesis 2. The higher the centralization of the internal advice network, the lower the degree of knowledge variety in a group.

Knowledge variety can also be enhanced through importing diverse knowledge from outside of the group. Studies on intergroup knowledge sharing have demonstrated that external ties are conduits of diverse and novel knowledge (e.g., Ancona & Caldwell, 1992; Hansen, 1999; Wong, 2004). Extending this perspective, this paper posits that the external network range of a group—that is, the number of advice ties to different groups—can impact knowledge variety by enhancing the group's capacity to absorb new knowledge into the group. Because people from external groups are likely to be exposed to different work experiences through either working on different tasks or interacting with different clients or individuals, a group with greater external network range is likely to be exposed to more diverse knowledge. Hence, at a minimum, groups involved in advice-seeking with more external groups are likely to develop greater knowledge variety because they have more opportunities to access diverse knowledge. In addition to access benefits, greater interaction with people of different worldviews is also likely to enhance members' capacity for absorbing new knowledge as they become more proficient at understanding worldviews that are different from their own. One important reason for knowledge exchange breakdowns is individuals' inability to interpret and understand the perspectives of people from diverse knowledge domains (Dougherty, 1992). Absorbing diverse knowledge requires the ability to reconcile dissimilarities across different knowledge domains (Krauss & Fussell, 1991). With greater external network range, the experience from greater contact with people of different worldviews can improve group members' ability at interpreting and appreciating diverse knowledge, thus facilitating the absorption of new knowledge. While Reagans and McEvily's (2003) study shows that greater network range increases the ease of knowledge transfer from a source to a recipient by increasing the source's ability to convey complex ideas, it is plausible that greater external network range can also increase the ability of group members to *absorb* diverse knowledge.

Hypothesis 3. The higher the range of the external advice network, the higher the degree of knowledge variety in a group.

Group knowledge influences on group effectiveness

Broadly, group effectiveness has been defined as the extent to which a group is able to meet its output goals and members' goals (Hackman, 1987; Shea & Guzzo, 1987). This study focuses on a group's effectiveness at accomplishing its tasks and, thus, examines group effectiveness as group managers' evaluations of task outputs in terms of efficiency, innovativeness, quality, and overall work excellence.

There is growing evidence from both research on new product development teams and experimental student teams that shared task knowledge improves group effectiveness (e.g., Hoopes & Postrel, 1999; Mathieu et al., 2000; Stasser & Titus, 1985; Stasser, Taylor, & Hanna, 1989). Hoopes and Postrel's (1999) study of new product development teams, for instance, illustrates that when groups have greater common task knowledge, coordination costs are reduced because shared knowledge reduces the likelihood of 'glitches' or errors arising from misunderstandings. Similarly, Dougherty's (1992) study of 'thought-worlds' in new product development suggests that individuals with different task knowledge domains can develop diverse perspectives and interpretations and, thus, become unable to communicate and coordinate their perspectives to produce more effective solutions. Greater shared knowledge,

through cross-training, for instance has been shown to enhance team performance through improving coordination for groups involved in highly interdependent and complex simulation tasks (Marks, Sabella, Burke, & Zaccaro, 2002). These studies suggest that for groups involved in complex tasks, greater task knowledge overlap can enhance group effectiveness because group members can better communicate and coordinate with one another, thereby reducing output errors and delays. Hence, for knowledge-intensive groups such as those examined in this research, where the task can be cognitively complex and requires members to leverage on one another's knowledge, we should also expect knowledge overlap to enhance group effectiveness through enabling members to coordinate effectively with one another to reduce errors and improve the quality of their solutions.

Hypothesis 4. The higher the knowledge overlap in a group, the higher the group effectiveness.

It has been posited that having a more varied repertoire of knowledge can enhance group effectiveness because a greater range of task knowledge enables the group to have more combinatorial possibilities to create novel solutions (Hargadon, 1999), to obtain a more accurate understanding of the environment (e.g., McNamara et al., 2002; Walsh & Fahey, 1986), and to apply different knowledge to changing task requirements (Lyles & Schwenk, 1992; Weick, 1995). For knowledge-intensive groups where the task is likely to involve complex problem solving, we can expect groups with greater knowledge variety to be able to recognize more environmental cues, consider a problem from more angles, and evaluate more solutions during problem solving. Consequently, such comprehensive problem solving is likely to lead to more creative and accurate solutions (Bantel & Jackson, 1989). As such, for knowledge-intensive groups, knowledge variety is likely to be positively related to group effectiveness.

Hypothesis 5. The higher the knowledge variety in a group, the higher the group effectiveness.

Group knowledge as mediator between advice network structures and group effectiveness

As previously highlighted, the key argument of the structural perspective of social capital is that the configuration of social relationships determines performance through affecting one's access to needed resources (Burt, 2000; Cummings & Cross, 2003). Specifically, how social relationships influence the knowledge available for group members to accomplish their task has often been cited as a key reason why social network structures enhance group performance (e.g., Cummings, 2004; Nahapiet & Ghoshal, 1998; Reagans & Zuckerman, 2001). As advice ties are characterized by the provision of information and ideas, we should expect the structure of advice ties both internal and external to a group to influence knowledge resources in a group and, thus, the effectiveness of the group at accomplishing its tasks. Specifically, it is proposed that advice network structures impact group effectiveness through the mediating effects of knowledge overlap and knowledge variety.

It has been argued in this paper that when group members are more interconnected in a dense internal advice network, members have greater opportunities to transfer what they know, thereby increasing the degree of shared task knowledge in a group. Because greater shared task knowledge can facilitate communication and coordination of information and ideas, it is likely to enhance group effectiveness by reducing misunderstandings and, thus, the extent of errors and delays (Hoopes & Postrel, 1999; Postrel, 2002). Hence, it is hypothesized that internal advice network density facilitates group effectiveness through developing higher shared task knowledge.

Hypothesis 6a. Internal network density is positively associated with group effectiveness.

Hypothesis 6b. Knowledge overlap mediates the relationship between internal network density and group effectiveness.

In the case of internal advice network centralization, it is hypothesized that greater centralization is likely to decrease group effectiveness through reducing the extent of knowledge variety in a group. Experimental research on communication networks conducted in the 1950s indicates that groups with more centralized communication networks performed worse at complex tasks (Shaw, 1964). Sparrowe et al.'s (2001) study of advice networks also found a negative relationship between advice network centralization and group performance. Similarly, given that knowledge-intensive groups tend to be involved in cognitively complex tasks, we should also expect internal network centralization to negatively impact group effectiveness. Because of the concentration of advice-seeking ties to a few members, the group has less opportunity to harness and combine the knowledge of different members to create new knowledge, thus leading to less knowledge variety. With less knowledge variety, the potential to create accurate and creative solutions is reduced, thereby leading to lower group effectiveness (e.g., Sutton & Hargadon, 1996). Hence, it is hypothesized that knowledge variety mediates the relationship between internal network centralization and group effectiveness.

Hypothesis 7a. Internal network centralization is negatively associated with group effectiveness.

Hypothesis 7b. Knowledge variety mediates the relationship between internal network centralization and group effectiveness.

In addition to internal advice ties, the importance of external advice ties to group effectiveness has also been highlighted in numerous studies of boundary-spanning in new product development teams (e.g., Ancona & Caldwell, 1992; Katz & Tushman, 1979). Because group effectiveness can depend on the synthesis of different ideas to create innovative solutions, advice ties with external groups—which are likely to have non-redundant knowledge—help to import diverse knowledge and, thus, increase the innovativeness of the group output (Ancona & Caldwell, 1992; Cummings, 2004; Hansen, 1999; Wong, 2004). Relatedly, as groups involved in knowledge-intensive tasks are also likely to be involved in complex problem solving, we should also expect groups with greater external network range to be more effective because they have greater knowledge variety to solve their problems. By forming advice ties to different external groups, members not only import diverse knowledge but also become more proficient at integrating the knowledge, thus building a wider range of knowledge that they can use to improve their group effectiveness. Thus, it is hypothesized that external network range is positively related to group effectiveness, and this relationship is mediated by knowledge variety.

Hypothesis 8a. External network range is positively associated with group effectiveness.

Hypothesis 8b. Knowledge variety mediates the relationship between external network range and group effectiveness.

Study 1

Study 1 was conducted where data were collected at two points in time from undergraduate students enrolled in a strategic management course at a large public university. The course spanned 12 calendar

weeks and was structured into 20 classes, each with about 20–25 students. As part of the course requirement, students were required to form groups of three to six members to participate in a computerized strategy simulation task. The simulation task required group members to form a firm to compete with another firm that is simulated by the computer program in a consumer bike industry over 10 decision periods, which corresponded to 10 calendar weeks. Each decision period is equivalent to a year in the simulated environment, and every group receives a performance report of their company after every decision period. Group members represent the top management team of their firm and would make different strategic decisions such as product and pricing decisions, manufacturing decisions, and new market development decisions. The primary objective of each group was to maximize the cumulative shareholder value of their company as a significant portion of each member's overall course grade is based on the cumulative shareholder value of their company at the end of the simulation task. Thus, all the groups were ongoing, bounded groups with shared responsibility for their group performance.

In total, 434 undergraduate students from 20 classes participated in this simulation task. On average, each class would have four to five groups. Students were assigned to the same class for the duration of the course and met at least once a week during class. As such, members of different groups in the same class can share their knowledge and experiences about how to improve their company performance. At the mid-point of the simulation task (i.e., after the fifth decision period), a Web-based questionnaire, which tapped at the task advice relationships within a group and between groups in the same class, was administered to all the group members. It seems reasonable to assess advice networks and group knowledge at the mid-point of the simulation task for two key reasons. First, sufficient time has lapsed for individuals to experience the task and form task advice ties with others. Second, previous research suggests that group members tend to reflect on and change task strategies at the mid-point (Gersick, 1988) and, thus, we can expect advice sought at this period to have a more significant impact on subsequent group performance.

A total of 92 groups participated in the simulation task but 12 groups were excluded from the final sample because only one member from these groups completed the questionnaire. Hence, the final sample available for analysis was 80 groups, yielding a response rate of 87 per cent. Group size ranged from three to six members (mean = 4), and the average response rate of each group was 96 per cent.

Measures

Internal advice network density and centralization

Respondents were asked to indicate for each individual in their class how often they turned to the individual for information and knowledge on the simulation task in the past 7 weeks based on a six-point Likert scale (0 = I do not know this person; 5 = Very frequently). A table listing of all individuals in the class was provided to each respondent. A $n \times n$ matrix (n = total number of individuals in a class) was created, where cell entry X_{ij} represents the frequency in which individual i seeks information and knowledge from individual j . The data are then dichotomized such that an advice tie exists if the response was greater than the scale mid-point of 3.

The internal advice network of each group was created by extracting only respondents' advice ties to other members in their group. *Internal network density* was measured by the ratio of actual to possible number of advice-seeking ties among members in the group. *Internal network centralization* was measured by Wasserman and Faust's (1994) measure of in-degree network centralization, which calculates the dispersion of or variance in individual actor's in-degree centrality indices across the

group. In-degree centrality of a group member in the internal advice network represents the total number of members who turn to the focal member for advice. Network centralization is computed as:

$$C = \frac{\left[\sum_{i=1}^g (C_D(n_i) - \bar{C}_D)^2 \right]}{g}$$

where C refers to in-degree network centralization of a group, $C_D(n_i)$ refers to in-degree centrality of member i , $\bar{C}_D$ refers to the mean member in-degree centrality, and g refers to the group size. In-degree centrality indices were used, as opposed to other centrality measures (e.g., betweenness or closeness centrality), because the focus of this study is on the equality of influence among members, as measured by whether members are equal givers of advice. The higher the in-degree centralization of the advice network, the more a single member is the object of advice-seeking, reflecting greater disparity in influence among the most central member and other members.

External advice network range

The external advice network of each group was created by extracting respondents' advice ties to non-group members in their class. As long as one member in a group has an advice tie to an individual from another group, the group is considered to have an external advice tie to that group. External network range of a group was measured by the number of distinct external groups upon which the group relies upon for knowledge. As the number of external groups a group can seek for advice depends in part on the number of groups in a class, this measure was standardized by dividing the number of external groups to which a group relies upon for knowledge by the total number of groups in the class.

Knowledge overlap and knowledge variety

As groups were involved in a computerized strategy simulation task, task-relevant knowledge domains in this case pertain to different business tactics respondents can use to increase the performance of their firm. Hence, respondents were asked to indicate the extent to which they have knowledge of how to implement different business tactics on a five-point Likert scale (1 = Not at all; 5 = To a very great extent). A list of 17 business tactics that was extracted from Miller, Burke, and Glick (1998) in their study on cognitive diversity in top management teams was provided to each respondent. Examples of business tactics include maintaining or seeking the lowest cost-per-unit in the industry, frequently developing new products/services, and offering a broad range of products/services.

Knowledge overlap in a group measures the extent to which group members have similar knowledge. Thus, the greater the variability in members' knowledge of different business tactics, the lower the degree of knowledge overlap in the group. Following Klein, Conn, Smith, and Sorra (2001) who used the average standard deviation statistic to measure cognitive variability in the group, I computed knowledge overlap by taking the reciprocal of the average standard deviation of group members' knowledge across the different business tactics. The average standard deviation of members' knowledge was calculated by first computing the standard deviation of members' knowledge in each business tactic and, then, averaging the standard deviation scores across the different business tactics.

Knowledge variety in a group measures the number of different task knowledge domains in the aggregated knowledge structure of the group (e.g., McNamara et al., 2002). Given that knowledge variety measures the coverage of a group in different knowledge domains, I averaged the members' knowledge in each business tactic to compute the group's knowledge in each tactic. A group is considered to have knowledge or expertise in a particular tactic if the group's average on that tactic is greater than the scale mid-point of 3, and is coded as '1' on that tactic. If not, the group is coded as '0'. Knowledge variety is measured by aggregating the '1s' across the different business tactics.

Group effectiveness

Group effectiveness was operationalized as cumulative shareholder value achieved by the firm of each group at the end of the simulation task. This was chosen as an indicator of the effectiveness of each group because students were specifically instructed at the beginning of the task to maximize this performance variable and a significant portion of the course grade was based on this variable.

Control variables

Group size, group ability, and task interdependency were included as control variables. Group size was included as a control variable because groups with larger size may have more varied and less overlapping knowledge. Group ability, measured by averaging group members' letter grades (1 = A; 2 = B; 3 = C, 4 = D, 5 = E; 6 = F) in the past 2 years, was also included as a control variable because groups with more able members may be able to make higher quality strategic decisions and, thus, perform better than those with less able members. The measure of group ability was multiplied by -1 so that a higher value reflects higher group ability. Task interdependency was measured by showing respondents a visual representation of team interdependency and asking them to indicate the percentage of time their group worked according to the visual representation. Group members' scores were averaged to form the task interdependency measure. Given that this is a single-item measure, the intraclass correlation coefficient (ICC) was computed to assess the degree of agreement within each group. The ICC was 0.54, and the one-way ANOVA from which the ICC was derived was significant at $p < 0.01$, thus supporting the aggregation of members' scores to the group level.

Results of Study 1

Table 1 reports the means, standard deviations, and intercorrelations of the variables. Hierarchical regression analyses were used to test all hypotheses, where control variables were entered in step 1, followed by the set of independent variables in step 2, and a significance test of the change in multiple squared correlation coefficient (R^2) from step 1 to step 2. Table 2 reports findings of the hierarchical regressions of knowledge overlap and knowledge variety on the control variables and network variables. Hypothesis 1 was supported as internal network density was positively related to knowledge overlap ($\beta = 0.21$, $p < 0.05$). Notably, even though external network range is not hypothesized to be associated with knowledge overlap, it was significantly and negatively associated with knowledge

Table 1. Study 1: correlation matrix and summary statistics ($N = 80$)

Variables	Mean	SD	1	2	3	4	5	6	7	8	9
1. Group size	4.34	0.84									
2. Group ability	-2.41	0.39	0.05								
3. Task interdependency	39.92	14.05	-0.06	0.02							
4. Internal network density	0.69	0.23	-0.10	0.08	0.17						
5. Internal network centralization	21.34	17.30	0.07	-0.01	-0.07	-0.32**					
6. External network range	0.32	0.27	0.21 ⁺	0.07	-0.08	0.03	-0.06				
7. Knowledge overlap	1.38	0.29	-0.26*	-0.02	0.12	0.18	0.08	-0.28*			
8. Knowledge variety	13.09	1.92	-0.08	-0.20 ⁺	0.09	0.15	-0.25*	0.33**	0.16		
9. Group effectiveness	809.99	452.40	-0.07	0.25*	0.32**	0.25*	0.11	0.13	0.22*	0.22*	

⁺ $p < 0.10$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

Table 2. Study 1: hierarchical regression results of knowledge overlap and knowledge variety^a

Variables	Knowledge overlap		Knowledge variety	
	Model 1	Model 2	Model 3	Model 4
Group size	−0.26*	−0.20 ⁺	−0.07	−0.12
Group ability	−0.01	−0.01	−0.20*	−0.23*
Task interdependency	0.10	0.06	0.09	0.09
Internal network density		0.21*		0.07
Internal network centralization		0.15 ⁺		−0.19*
External network range		−0.23*		0.36***
R^2 (adjusted R^2)	0.08 (0.04)	0.17 (0.10)	0.05 (0.01)	0.24 (0.18)
ΔR^2		0.09*		0.19**
ΔF		2.73		6.02

Standardized coefficients are reported.

^{ca}As all hypotheses were directional, one-tailed tests of statistical significance were used.

^{a+} $p < 0.10$; ^{b*} $p < 0.05$; ^{c**} $p < 0.01$; ^{d***} $p < 0.001$.

overlap ($\beta = -0.23$, $p < 0.05$). Hypothesis 2 was supported as internal network centralization was negatively and significantly associated with knowledge variety ($\beta = -0.19$, $p < 0.05$). Hypothesis 3 was also supported as external network range was positively associated with knowledge variety ($\beta = 0.36$, $p < 0.001$).

Table 3 reports findings of the hierarchical regression of group effectiveness on the control, network, and group knowledge variables. Consistent with Hypotheses 6a and 8a, internal network density ($\beta = 0.22$, $p < 0.05$) and external network range ($\beta = 0.17$, $p < 0.05$) were positively and significantly associated with group effectiveness. Thus, Hypotheses 6a and 8a were supported. Surprisingly, contrary to Hypothesis 7a, internal network centralization was positively and significantly related to group effectiveness ($\beta = 0.23$, $p < 0.05$). When knowledge overlap and variety were added to the regression model, knowledge variety was positively and significantly associated with group effectiveness ($\beta = 0.24$, $p < 0.05$), thus supporting Hypothesis 5. However, Hypothesis 4 was not supported as knowledge overlap was not significantly associated with group effectiveness ($\beta = 0.13$, $p > 0.10$).

Although Baron and Kenny's (1986) mediation test has been widely used in management research, it has recently been criticized to have low statistical power to detect mediation when it exists in the

Table 3. Study 1: hierarchical regression results of group effectiveness^a

Variables	Model 1	Model 2	Model 3	Model 4	Model 5
Group size	−0.05	−0.07	−0.02	−0.04	−0.04
Group ability	0.24*	0.21*	0.28**	0.22*	0.28**
Task interdependency	0.30**	0.29**	0.26**	0.28**	0.27**
Internal network density		0.22*	0.18 ⁺	0.18*	0.20*
Internal network centralization		0.23*	0.25*	0.20*	0.28**
External network range		0.17*	0.12	0.21*	0.08
Knowledge variety			0.24*		0.27*
Knowledge overlap			0.13	0.19*	
R^2 (adjusted R^2)	0.16 (0.12)	0.25 (0.19)	0.32 (0.24)	0.28 (0.21)	0.31 (0.24)
ΔR^2		0.09*	0.07*	0.03 ⁺	0.06*
ΔF		3.06	3.51	2.92	5.70

Standardized coefficients are reported.

^{aa}As all hypotheses were directional, one-tailed tests of statistical significance were used.

^{b+} $p < 0.10$; ^{c*} $p < 0.05$; ^{d**} $p < 0.01$; ^{***} $p < 0.001$.

population (Mackinnon, Lockwood, Hoffman, West, & Sheets, 2002). Hence, this paper examines the hypothesized mediating effects using the bootstrap methodology (Shrout & Bolger, 2002) and the product of coefficients tests (Mackinnon et al., 2002). The bootstrap methodology is useful to detect indirect effects when sample sizes are moderate or small (Shrout & Bolger, 2002), and the product of coefficients tests have been shown, through simulation studies, to have the most statistical power to detect indirect effects (Mackinnon et al., 2002).

Essentially, the bootstrap method generates an actual distribution of the estimated indirect effect by sampling with replacement from an existing data set, thereby generating multiple data sets from which estimates of the indirect effect can be obtained. As there is an actual distribution of the estimated indirect effect, the 95 per cent confidence interval for the indirect effect is not estimated using normal theory, which can reduce the statistical power to detect mediation (Mackinnon et al., 2002; Shrout & Bolger, 2002). In this paper, AMOS, a structural equation modeling package, is used to conduct the bootstrap method. Following Mackinnon et al.'s (2002) recommendation, two products of coefficients tests, specifically the distribution of products method ($P = z_{\alpha}z_{\beta}$) and distribution of $\alpha\beta/\sigma_{\alpha\beta}$ method ($z' = \alpha\beta$), are also used to test the hypothesized mediating effects. α represents the relation between the independent and mediating variable and β represents the relation between the mediating and dependent variables adjusted for the effect of the independent variable. In order to establish a mediating effect, both paths α and β should be significantly greater than zero, and either the P statistic or the z' statistic should be significant. As the P and z' statistics are not normally distributed, the critical value to determine 0.05 significance level is 2.18 and 0.97 for the P and z' statistics, respectively (Mackinnon et al., 2002).

Results for the bootstrap method indicate that the indirect effect of internal network density on group effectiveness through knowledge overlap was not significant, as the bias-corrected 95 per cent confidence interval for this indirect effect ranged from -15.845 to 295.267 ($p = 0.136$). For the product of coefficients tests, although both the P and z' statistics for this indirect effect were significant ($P = 2.48$, $z' = 1.09$, $p < 0.05$), regression findings presented earlier indicated that knowledge overlap was not significantly related to group effectiveness (i.e. $\beta = 0$). Hence, Hypothesis 6b was not supported.

In the case of Hypothesis 7b, the indirect effect of internal network centralization on group effectiveness through knowledge variety approached significance, as the bias-corrected 95 per cent confidence interval for this indirect effect ranged from -5.289 to 0.078 ($p = 0.074$). However, results from the product of coefficients tests show that both the P and z' statistics ($P = -4.18$, $z' = -1.41$, $p < 0.05$) were significant. Regression findings presented earlier also indicated that α and β were jointly significant, as internal network centralization was significantly associated with knowledge variety, and knowledge variety was significantly associated with group effectiveness. However, given the mixed findings, Hypothesis 7b was not supported.

Hypothesis 8b, which posits knowledge variety as the mediator of the relationship between external network range and group effectiveness, was supported as the bias-corrected 95 per cent confidence interval for this indirect effect ranged from 20.878 to 399.429 ($p = 0.026$). Both the P and z' statistics ($P = 8.27$, $z' = 1.97$, $p < 0.05$) were significant, and the test of joint significance of α and β was also satisfied as previously presented regression results show that external network range was significantly related to knowledge variety, and knowledge variety was significantly related to group effectiveness.

As a test of added robustness, the Baron and Kenny's (1986) mediation test was also conducted. Results for Hypothesis 6b and 8b are consistent with those tested using the bootstrap and product of coefficients test. Results from Baron and Kenny's test also did not support Hypothesis 7b. Internal network centralization was positively and significantly associated with group effectiveness ($\beta = 0.23$, $p < 0.05$) but when the knowledge variables were added to the regression model, internal network centralization remained significantly associated with group effectiveness ($\beta = 0.25$, $p < 0.05$).

Study 2

Study 2 consists of work groups from three firms—a hospital, an industrial diversified firm, and a high-technology firm. A diverse array of permanent work groups were surveyed, including nursing and physician groups from the hospital, engineering groups from the industrial diversified firm, and marketing and sales groups from the high-technology firm. Despite diversity in the specific types of work the groups undertake, preliminary interviews with the group managers from three firms revealed that the groups are similar in the relatively high knowledge-intensive nature of their work. All work groups across the three firms have to acquire and leverage different knowledge to solve challenging problems.

Group network data were collected from 50 work groups from three firms—a hospital ($N = 18$), an industrial diversified firm ($N = 19$), and a high-technology firm ($N = 13$). Data were collected through Web-based questionnaires administered to all members of each work group. A criterion of at least a 66.67 per cent response rate led to the dropping of 10 work groups. Hence, the final sample available for analysis was 40 organizational work groups, of which 15 were from the hospital, 13 from the industrial diversified firm, and 12 from the high-technology firm. The sample response rate was 80 per cent and, on average, the group response rate in the sample was 86.8 per cent. Group size ranged from 3 to 25 members (mean = 9.9), and group tenure ranged from 0.3 to 16.4 years (mean = 3.4 years).

Measures

Internal advice network density and centralization

The internal advice network was measured by asking each respondent to indicate the members in his or her workgroup whom (1) ‘you usually go to for advice in your work.’ A name generator method was used whereby no restrictions were placed on the number of nominations so as to reduce measurement error. In contrast to Study 1 where valued data (i.e., frequency with which member turns to others for task advice) were collected, time constraints of respondents in Study 2 precluded the collection of valued data. As the computation of network density and network centralization indices requires a square matrix, I excluded from the network analyses those group members who were nominated by other members but did not respond to the questionnaire. Similar to Study 1, *internal network density* was measured by the ratio of actual to possible number of advice-seeking ties among members in the group; and *internal network centralization* was measured by Wasserman and Faust’s (1994) measure of in-degree network centralization.

External advice network range

The external advice network of the work group was measured by asking group members to list ‘other groups from whom you obtain ideas/advice/information for your group’s work.’ I compared the external group lists generated by all the group members to create an overall list of the distinct groups that their group relies upon for advice. External network range of a group was measured by the number of distinct external groups upon which the group relies upon for knowledge.

Knowledge overlap and knowledge variety

Following Rulke and Galaskiewicz (2000) where distribution of knowledge in groups was measured by members’ ratings of their knowledge on functional areas (e.g., marketing, production, etc), knowledge overlap and knowledge variety were derived from group members’ ratings of their level of knowledge

(1 = not at all, 5 = to a very great extent) across different domains of task knowledge. Given that work groups from three different firms were sampled, I created a unique list of broad domains of task knowledge that were applicable to work groups in each firm.

Preliminary interviews were conducted with two group managers from each firm to create a list of task knowledge that was relevant to the task performance of the groups from each firm. In order to ensure that the list created is representative of the task-relevant knowledge domains required in each group, I surveyed group managers of each workgroup on the relevance of knowledge of each task, on a scale from 1 (not at all job relevant) to 5 (extremely job relevant), to the tasks of the group. On average, group managers from the hospital, industrial diversified firm, and high-technology firm rated the relevance of the list of task knowledge as 4.2, 3.3, and 3.5, respectively. Although I included two 'Other' open-ended categories so that group managers could input additional task knowledge that might not be included in the given list, no additional task knowledge domains were suggested.

For groups from both the diversified industrial and high-technology firm, 13 task knowledge domains were created. The task knowledge domains for the groups from the diversified industrial firm include mechanical engineering knowledge, electrical engineering knowledge, operations management knowledge, and quality control knowledge. The task knowledge domains for groups from the high-technology firm include networking technology, wireless technology, product design knowledge, and marketing or sales knowledge. For groups from the hospital, 11 task knowledge domains were used to capture members' task knowledge, including physician knowledge, medical technology knowledge, nursing knowledge/skills, knowledge of regulations or legal requirements, and knowledge of clinical guidelines and protocols. As in the manager-level survey, I created two 'Other' open-ended categories for group members to input additional task knowledge domains that might not be included in the list. The 'Other' task knowledge domains were included to compute the collective knowledge variables only if at least two members from the same group suggested the same 'Other' knowledge domain, because having at least one other member nominate the same knowledge domain helps to verify that the knowledge is relevant to the group task. Of the 40 workgroups, only 28 out of 300 members who responded to the questionnaire nominated at least one 'Other' task knowledge domain, and the two-member criteria was not met in any of these cases. Hence, I did not use the 'Other' task knowledge domains to compute the group knowledge variables.

Following Study 1, *knowledge overlap* was computed by taking the reciprocal of the average standard deviation of group members' knowledge across different task knowledge domains. The measure of *knowledge variety* also followed Study 1, where a group is considered to have expertise of a particular knowledge domain if the group average in that domain is greater than 3 (and, thus, coded as '1' in that domain). If not, the group is coded as '0'. The degree of knowledge variety in a group is then measured by aggregating the '1s' across the different knowledge domains. I standardized the measure by dividing the aggregated score of each group by the total number of task knowledge domains surveyed.

Group effectiveness

Group effectiveness was measured by asking group managers to evaluate the group in terms of efficiency, innovativeness, quality, and work excellence, on a five-point Likert scale (1 = poor; 5 = superior). The group managers were not actively involved in the day-to-day operations of the groups, but the individuals were formally designated by the firm to evaluate group performance. As such, the risk of bias is minimized as these managers are less likely to cloud the performance evaluation of their group based on relational factors that may have developed through close interactions with group members. Each manager's ratings across the four dimensions were averaged to form an overall group effectiveness measure ($\alpha = 0.85$).

Control variables

Given that work groups from different firms were sampled, I included firm-level dummy variables to control for differences in knowledge overlap and variety that could be due to firm-specific work that the groups engage in. For example, hospital work groups may have greater knowledge variety because multidisciplinary groups of doctors and nurses are often formed in the delivery of medical care. I also included group demographics such as size, tenure, and educational heterogeneity as control variables. We can expect groups with larger size or lower group tenure to have more varied and less overlapping knowledge because having a bigger group increases the probability that group members have different knowledge bases, and group members who have had less time to share knowledge would be more differentiated in their knowledge bases. Similarly, groups with greater educational heterogeneity are also likely to have less knowledge overlap because members are less likely to interact with one another. Group size was measured by the number of members in a group, and group tenure was measured by the average length of time each member had been a member of his/her group. Educational background was measured by asking respondents to indicate their primary educational background (e.g., accounting, engineering, finance, sciences, etc), and the Blau's (1977) heterogeneity index was used to compute educational heterogeneity.

As the sample consists of work groups involved in different types of task, task type and task interdependency were also included as control variables. Task type was measured by examining the proportion of time a group spends on conceptual tasks versus procedural tasks (e.g., Goodman, 1986). We can expect groups that are involved in greater conceptual tasks to have greater opportunities to share knowledge and, thus, have greater knowledge overlap and variety. Following Stewart and Barrick (2000), I measured task type by asking respondents to indicate the percentage of time their group spends on generating ideas and plans, choosing alternatives, negotiating conflicts, and executing work. I used the group mean on the total percentage of time spent on the first three categories to indicate the degree to which a group's work is conceptually oriented. Given that these items are formative rather than reflective, I conducted separate ICC for each item to assess the extent of agreement within groups. The ICC scores ranged from 0.25 to 0.76, and the one-way ANOVA tests from which the ICCs were derived were significant at $p < 0.01$ except for the negotiating conflicts item which tended to significance at $p = 0.08$. Thus, overall, there was high within group agreement which supported the aggregation of members' responses to the group level.

Task interdependency was measured by the shortened version of Kiggundu's (1983) task interdependence scale ($\alpha = 0.85$). Depending on the degree of task interdependency, group members can have varying opportunities to interact, which can affect the extent of knowledge overlap and variety in the group. Since task interdependency is measured by multiple items, the r_{wg} (George & James, 1993) was computed for each group. The average r_{wg} was 0.89, which indicates a high level of agreement within groups and, thus, supports the aggregation of members' responses to form a group task interdependency score.

Results of Study 2

Table 4 reports the means, standard deviations, intercorrelations, and Cronbach alphas of the variables. Given that work groups from different firms were included in the sample, I conducted a multivariate analysis of variance to examine the degree to which the groups were comparable. Results indicate that there was no significant difference in group demographics and network structures in the work groups from the three firms. Hence, I pooled the data for all subsequent analyses.

Table 4. Study 2: correlation matrix and summary statistics^a ($N = 40$)

Variables	Mean	SD	1	2	3	4	5	6	7	8	9	10	11	12	13
1. Firm dummy 1	0.38	0.49	n.a.												
2. Firm dummy 2	0.33	0.47	-0.54***	n.a.											
3. Group size	9.93	5.38	-0.16	0.10	n.a.										
4. Group tenure	3.41	3.65	0.14	0.18	0.25	n.a.									
5. Task type	0.500	0.17	0.09	-0.26	-0.25	-0.53***	n.a.								
6. Educational heterogeneity	0.42	0.23	0.02	-0.36*	0.16	0.08	0.22	n.a.							
7. Task interdependency	3.29	0.47	0.16	-0.07	-0.08	0.28 ⁺	0.07	0.09	(0.85)						
8. External network range	12.20	4.71	0.04	0.15	0.47**	0.30 ⁺	-0.06	0.22	0.15	n.a.					
9. Internal network density	0.44	0.21	0.02	-0.05	-0.67***	-0.04	0.19	-0.19	0.33*	-0.38*	n.a.				
10. Internal network centralization	49.86	18.33	0.03	0.03	-0.03	-0.14	0.07	-0.03	-0.02	-0.07	-0.11	n.a.			
11. Knowledge overlap	1.19	0.21	-0.27 ⁺	0.15	-0.43**	-0.32*	0.21	-0.37*	-0.20	-0.45**	0.64***	-0.10	n.a.		
12. Knowledge variety	0.66	0.20	0.22	-0.40*	-0.17	-0.14	0.32*	0.14	0.19	0.25	0.20	-0.25	0.02	n.a.	
13. Group effectiveness	3.76	0.72	0.21	-0.07	-0.16	-0.03	-0.10	-0.14	0.13	0.17	0.21	-0.05	-0.09	0.49**	(0.85)

^aCronbach's alphas are listed on the diagonal in parentheses.⁺ $p < 0.10$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

Table 5. Study 2: hierarchical regression results of knowledge overlap and knowledge variety^a

Variables	Knowledge overlap		Knowledge variety	
	Model 1	Model 2	Model 3	Model 4
Firm dummy 1	−0.40*	−0.15	−0.04	−0.02
Firm dummy 2	−0.13	0.04	−0.36 ⁺	−0.44*
Group size	−0.39**	0.03	−0.07	−0.21
Group tenure	0.07	−0.12	0.02	−0.22
Educational heterogeneity	−0.39**	−0.23 ⁺	−0.04	−0.03
Task type	0.25	0.15	0.22	0.04
Task interdependency	−0.18	−0.30*	0.15	0.09
Internal network density		0.57***		0.12
Internal network centralization		−0.14		−0.32*
External network range		−0.14		0.47**
<i>R</i> ² (adjusted <i>R</i> ²)	0.47 (0.35)	0.67 (0.56)	0.24 (0.07)	0.49 (0.31)
ΔR^2		0.20**		0.25**
ΔF		6.02		4.65

Standardized coefficients are reported.

^aAs all hypotheses were directional, one-tailed tests of statistical significance were used.

⁺*p* < 0.10; **p* < 0.05; ***p* < 0.01; ****p* < 0.001.

Table 5 shows results of the hierarchical regressions of knowledge overlap and knowledge variety on the control variables, and network variables. Consistent with Study 1, Hypotheses 1, 2, and 3 were supported as internal network density was positively related to knowledge overlap ($\beta = 0.57$, $p < 0.001$), internal network centralization was negatively associated with knowledge variety ($\beta = -0.32$, $p < 0.05$), and external network range was positively related to knowledge variety ($\beta = 0.47$, $p < 0.01$).

Table 6 reports findings of the hierarchical regression of group effectiveness on the control variables, network variables, and group knowledge variables. Although internal network density was positively

Table 6. Study 2: hierarchical regression results of group effectiveness^a

Variables	Model 1	Model 2	Model 3	Model 4
Firm dummy 1	0.23	0.26	0.21	0.27 ⁺
Firm dummy 2	0.02	−0.02	0.23	0.22
Group size	−0.11	−0.08	0.03	0.03
Group tenure	−0.21	−0.42 ⁺	−0.35 ⁺	−0.30 ⁺
Educational heterogeneity	−0.06	−0.03	−0.10	−0.02
Task type	−0.25	−0.42 ⁺	−0.39*	−0.44*
Task interdependency	0.17	0.04	−0.12	−0.01
Internal network density		0.36 ⁺	0.51*	0.29 ⁺
Internal network centralization		−0.07	0.04	0.10
External network range		0.45*	0.14	0.19
Knowledge variety			0.53**	0.54**
Knowledge overlap			−0.38 ⁺	
<i>R</i> ² (adjusted <i>R</i> ²)	0.12 (0.00)	0.30 (0.05)	0.49 (0.27)	0.45 (0.23)
ΔR^2		0.17 ⁺	0.19*	0.15*
ΔF		2.35	5.23	7.58

Standardized coefficients are reported.

^aAs all hypotheses were directional, one-tailed tests of statistical significance were used.

⁺*p* < 0.10; **p* < 0.05; ***p* < 0.01; ****p* < 0.001.

related to group effectiveness, this relationship only approached significance ($\beta = 0.36$, $p < 0.10$). Thus, Hypothesis 6a was not supported. Similarly, Hypothesis 7a was not supported as internal network centralization was not significantly related to group effectiveness ($\beta = -0.07$, $p > 0.10$). Similar to findings in Study 1, external network range was positively and significantly associated with group effectiveness ($\beta = 0.45$, $p < 0.05$), thus supporting Hypothesis 8a. As in Study 1, Hypothesis 4 was not supported because knowledge overlap was not significantly associated with group effectiveness ($\beta = -0.38$, $p < 0.10$). Hypothesis 5 was supported, as knowledge variety was positively and significantly associated with group effectiveness ($\beta = 0.53$, $p < 0.01$).

Similar to Study 1, the bootstrap method and product of coefficients tests were used to examine Hypotheses 6b, 7b, and 8b. The indirect effect of internal network density on group effectiveness through knowledge overlap approached significance, as the bias-corrected 95 per cent confidence interval for this indirect effect ranged from -2.662 to 0.265 ($p = 0.095$). Although the P and z' statistics for this indirect effect were significant ($P = -5.42$, $z' = -1.43$, $p < 0.05$), the regression findings presented have shown that knowledge overlap was not significantly related to group effectiveness (i.e., $\beta = 0$). Hence, Hypothesis 6b was not supported.

Results from the bootstrap method show that the indirect effect of internal network centralization and group effectiveness through knowledge variety approached significance, as the bias-corrected 95 per cent confidence interval for this indirect effect ranged from -0.015 to 0.001 ($p = 0.076$). In the case of the product of coefficients tests, both the P and z' statistics were significant ($P = -4.13$, $z' = -1.32$, $p < 0.05$). Regression findings also demonstrated that α and β were jointly significant, as internal network centralization was significantly associated with knowledge variety, and knowledge variety was significantly associated with group effectiveness. However, as with Study 1, Hypothesis 7b was not supported because of the mixed findings.

Hypothesis 8b was supported as the bias-corrected 95 per cent confidence interval for the indirect effect of external network range on group effectiveness through knowledge variety ranged from 0.002 to 0.081 ($p = 0.03$). Both the P and z' statistics were significant ($P = 7.93$, $z' = 1.98$, $p < 0.05$), and α and β were also shown in the regression results to be jointly significant as external network range was significantly related to knowledge variety, and knowledge variety was significantly related to group effectiveness. Hence, the indirect effect of external network range on group effectiveness through knowledge variety was supported.

As with Study 1, Baron and Kenny's mediation method was also conducted to test the consistency of results. Again, findings for Hypotheses 6b and 8b are consistent with those tested using the bootstrap and product of coefficients method. Results from Baron and Kenny's test also did not support Hypothesis 7b, as internal network centralization was not significantly associated with group effectiveness ($\beta = -0.07$, $p > 0.10$). The implications of these findings are discussed in the next section.

Discussion

In recent years, we have seen growing academic and practitioner interest in the roles of advice networks and group knowledge on group effectiveness (Cummings, 2004; Hansen, 1999; Reagans & McEvily, 2003). This research extends current understanding in this area in three ways. First, drawing from social capital theory, this research offers a framework to predict how advice network structures can be leveraged to optimize knowledge variety and knowledge overlap in groups. Consistent with implicit assumptions, internal network density and external network range are positively related to knowledge overlap and knowledge variety respectively. This research also adds to existing theory by showing that

knowledge variety and knowledge overlap are also a function of the internal and external advice network, respectively. Specifically, greater internal network centralization is associated with less knowledge variety, and there is suggestive evidence that greater external network range is associated with less knowledge overlap. Second, this research disentangles the relationships among knowledge variety, knowledge overlap, and group effectiveness by carefully distinguishing variety and overlap as two distinct knowledge dimensions. Findings show that task knowledge variety, but not task knowledge overlap, is positively associated with group effectiveness. This highlights the importance of knowledge breadth to group effectiveness, which has received relatively little theoretical and empirical attention compared to work on shared task knowledge. Third, this research directly tests whether group knowledge mediates the effects of advice network structures on group effectiveness. Results indicate that knowledge variety mediates the relationship of external network range and group effectiveness.

Theoretical implications

Both studies provide clear evidence that a dense internal advice network creates shared task knowledge and a large external advice network fosters varied task knowledge, thus providing empirical validation for the implicit understanding that networks transfer and create knowledge. In addition, findings from both studies also indicate that the internal advice network—specifically, the degree of centralization—can impact knowledge variety. Greater advice network centralization is associated with lower knowledge variety, suggesting that unequal member influence can hinder knowledge creation and, thus, result in lower knowledge variety.

That different aspect of the internal advice network affects knowledge variety and overlap portends that different internal social forces in a group drive knowledge creation and knowledge transfer. Burt (1980) indicated that network density and network centralization capture two distinct and critical dimensions of a network, with the former indicating the *intensity of relations* in a group and the latter indicating the *status ordering* in a group. Hence, a dense network is likely to foster direct interpersonal access and communication, while a centralized network is likely to lead to unequal member influence. To the extent that knowledge overlap is associated with network density and knowledge variety is associated with network centralization, we can infer that while knowledge transfer is driven by direct access and communication, knowledge creation is driven by mutual influence in task advice-giving. Hence, different social forces may be at work in affecting group members' propensity to *transfer* knowledge and propensity to *create* knowledge.

A surprising finding of Study 1 is the negative association between external network range and knowledge overlap, suggesting that a larger external network can trigger greater divergence in task knowledge in a group. It is plausible that groups whose members interact with a larger set of external contacts develop knowledge that become more similar to that of their external contacts and, consequently, become more divergent in knowledge with other group members. To the extent that greater knowledge overlap can facilitate group performance in certain types of groups, it would be important for future research to consider the potential trade-offs of external networks (i.e., increase knowledge variety but reduce knowledge overlap) in promoting group effectiveness.

Distinguishing between task knowledge variety and overlap, this research also examines their independent associations with group effectiveness. Both studies indicate a positive association between task knowledge variety and group effectiveness. For groups involved in knowledge-intensive work (e.g., physician and nursing groups require different clinical knowledge, and marketing groups require knowledge of different environmental trends and promotion techniques), a greater variety of task knowledge can enhance the group's ability to recognize and deal with multiple issues in its environment and, thus, increase the quality of its task solutions (Lyles & Schwenk, 1992; Weick, 1995). Surprisingly,

findings from both studies show a non-significant relationship between knowledge overlap and group effectiveness. This contrasts with studies of new product development teams (Hoopes & Postrel, 1999) and experimental student teams (Mathieu et al., 2000) that have found a positive relationship between knowledge overlap and group effectiveness. Perhaps, the high intensity of time pressure for the new product development and experimental teams in these studies accentuated the coordination benefits of knowledge overlap and, consequently, increase the importance of knowledge overlap for team performance. In contrast, the student groups in Study 1 had a 10-week period to plan their decisions; and the organizational workgroups—as permanent groups—in Study 2 may have developed other kinds of knowledge such as group process or group member knowledge to coordinate effectively with one another such that the coordination benefits of shared task knowledge are diminished. Another plausible reason could be that new product development teams have a higher level of task complexity or uncertainty than the groups examined in this research, such that task knowledge overlap is more important for those teams.

Overall, the significant association of task knowledge variety, but not task knowledge overlap, with group effectiveness is noteworthy, particularly because of the relatively higher theoretical attention that has been devoted to shared knowledge. Although the findings in this study suggest that knowledge breadth is more important to group effectiveness than knowledge overlap, I am hesitant to conclude that knowledge-intensive groups should focus solely on accumulating a wide range of knowledge without considering the degree of knowledge redundancy in the group. Instead, this research underscores the need for future research on group knowledge to more carefully distinguish the effects of knowledge variety and overlap in different organizational settings (e.g., temporary vs. permanent groups, time-pressured situations, task complexity, etc.) so as to provide greater theoretical and empirical clarity on how and when breadth of knowledge and shared knowledge affect group success.

Through tapping directly into the knowledge dimensions, this study is also able to parse apart the influence of advice networks and group knowledge on group effectiveness, thus testing whether advice networks impact group effectiveness through group knowledge. There is clear evidence from both studies that knowledge variety mediates the positive relationship between external networks and group effectiveness. Interestingly, while internal network centralization is positively associated with group effectiveness in Study 1, it is not significantly associated with group effectiveness in Study 2. The positive direct effect of network centralization in Study 1 contrasts with Sparrowe, Liden, Wayne and Kraimer's (2001) field study, which found a weak and negative association between advice network centralization and group performance. Perhaps, because student groups do not work on a daily basis on the task, more advice network centralization aids group performance as knowledge about the task is communicated more efficiently. Overall, mediation tests from both studies also indicate that knowledge variety did not mediate the relationship between internal network centralization and group effectiveness. Thus, even though internal network centralization is associated with knowledge variety, its impact on group effectiveness—as in the case of Study 1—is not due to its effect on knowledge variety. Across both studies, there is also suggestive evidence that internal advice network density positively influences group effectiveness independently of the degree of shared knowledge in a group. It is plausible that even though advice networks are instrumental networks, as opposed to expressive networks such as friendship networks, when more group members are involved in the exchange of task knowledge, groups may develop relational resources such as social cohesion that promote group effectiveness.

Managerial implications

The strong positive relationship between task knowledge variety and group effectiveness highlights to managers the value of promoting knowledge breadth for groups involved in knowledge-intensive work.

As the need to recognize, interpret, and adapt to multiple cues in the external environment is likely to be accentuated in knowledge-intensive work, managers need to ensure that their groups have a wide range of task knowledge available to meet the demands of their environment. Particularly, this study demonstrates the value of both internal and external advice networks in building knowledge variety in groups. Fostering advice-seeking ties to different individuals, both within and outside of the group, is likely to promote knowledge variety. However, recognizing that group members' time is scarce and building (and maintaining) both internal and external networks are time consuming, managers can institutionalize 'networking' roles such that the group as a whole has non-redundant external ties. For instance, group managers can appoint members as liaisons to different external groups so that the group as a whole has a wider external network. Managers can also confer different expert roles on each member so as to encourage advice-seeking among more members and foster a less centralized advice network.

Limitations and future directions

Although this study investigated the implied associations between advice network structures and group knowledge, I did not examine more proximal intervening mechanisms such as knowledge transfer and creation processes that relate network structures to group knowledge. To further advance our understanding, it would be beneficial for future studies to examine these intervening processes. In addition, as the effect of advice network density on group effectiveness is independent of shared task knowledge, we need more studies that investigate the causal mechanisms through which dense networks impact performance.

Another possible limitation of this study is the use of self-reported knowledge to compute knowledge overlap and knowledge variety. Although the use of self-reported knowledge is consistent with studies on knowledge in groups (e.g., Rulke & Galaskiewicz, 2000), there is a risk of self-inflation, as group members may rate themselves as having higher knowledge than is actually the case. Nonetheless, to the extent that the risk of self-inflation affects all group members, the knowledge overlap measure is not adversely affected, as it measures variance across the group, and not amount of knowledge. The knowledge variety measure, however, could be artificially inflated. Even if this is the case, there is no *a priori* reason to assume that the structural network predictors would also be artificially inflated such that the observed associations between the network predictors and group knowledge variables are due solely to artificial covariance (Podsakoff & Organ, 1986). Regardless, future studies should collect more objective measures of knowledge in order to obtain more accurate measures of group knowledge.

Although Study 2 is a cross-sectional study, Study 1 measured group effectiveness after a time lag from the assessment of the independent variables, thus allowing a test of causality of advice networks and group knowledge on group effectiveness. However, ideally, advice networks and group knowledge should also be measured at two separate points in time so that their causal relationships can be assessed. Particularly, it is plausible that the causal relationship between network density and knowledge overlap is reversed, since the homophily argument predicts that more similar individuals engage in more communication and, thus, foster greater network density. In mitigation, the inclusion of educational heterogeneity as a control in Study 2 helps to partial out the effect posited by the homophily argument, as we can expect less educational heterogeneous groups (i.e., greater similarity) to have higher network density. The reverse causation explanation may also be more applicable in studies that examine expressive social networks, rather than advice networks, as it is equally reasonable to expect group members with high shared knowledge to have less dense advice-seeking networks since they do not need to seek the same types of knowledge. Regardless, a longitudinal study that teases out the

reciprocal causal relationships between network structures and group knowledge would be a fruitful area for future research.

A strength of this research is the use of a two-study approach to investigate the proposed theoretical framework. While Study 1 provides some test of causality of the impacts of advice networks and group knowledge on group effectiveness, Study 2 presents a test of generalizability of the findings in organizational work groups. Consequently, this research offers a more nuanced understanding of how advice networks and group knowledge affect group effectiveness. Overall, the pattern of findings from both studies indicates the value of using a social structural network approach to examining group knowledge; and also highlights that while the value of certain networks can be attributed to their knowledge benefits, as in the case of external network range, this may not be true in other cases, such as internal network density.

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